# Palatial Pools Pty Ltd
## Pool Area Shield — Proprietary Technology
### Technical Documentation and Commercial Licensing Overview
**Classification:** PROPRIETARY & CONFIDENTIAL
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## 1. Introduction

The **Pool Area Shield** (PAS) is an engineered technology platform developed by Palatial Pool Services Pty Ltd to provide enhanced safety monitoring solutions for commercial and residential aquatic environments. It is designed as a supplementary safety system that leverages advanced sensor fusion, real-time signal processing, and secure data communications to detect and alert on potential pool area intrusions or unsafe conditions.

This document provides a comprehensive technical overview of the Pool Area Shield technology, including architectural design, algorithmic foundations, data handling protocols, and integration specifications. It is intended for engineers, property managers, safety regulators, and commercial licensees evaluating or deploying the system.

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## 2. System Architecture

The Pool Area Shield system comprises three primary layers:

- **Sensing Layer:** Distributed sensor nodes deployed around the pool perimeter to detect presence and motion.
- **Processing Layer:** Edge and cloud-based signal processing units that analyze sensor data to identify safety events.
- **Interface Layer:** User-facing dashboards, alerting systems, and API endpoints for monitoring and integration.

### High-Level Architecture Diagram

```mermaid
graph LR
    S1[Sensor Node 1] -->|Wireless Data| EP1[Edge Processor]
    S2[Sensor Node 2] -->|Wireless Data| EP1
    S3[Sensor Node N] -->|Wireless Data| EP1
    EP1 -->|Encrypted Link (TLS 1.3)| CloudServer[Cloud Processing & Storage]
    CloudServer -->|API/SDK| Dashboard[Monitoring Dashboard]
    CloudServer -->|Alerts| NotificationSystem[SMS/Email/Push Alerts]
```

- **Sensor Nodes:** Utilize a combination of infrared, ultrasonic, and optical sensors to detect human presence and movement near the pool area.
- **Edge Processor:** Performs low-latency preprocessing, noise filtering, and preliminary event detection to reduce cloud load.
- **Cloud Server:** Runs advanced machine learning models, event correlation, long-term data storage, and provides interface endpoints.
- **Dashboard & Notifications:** Enables real-time monitoring, historical data review, and alert configuration for authorized users.

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```

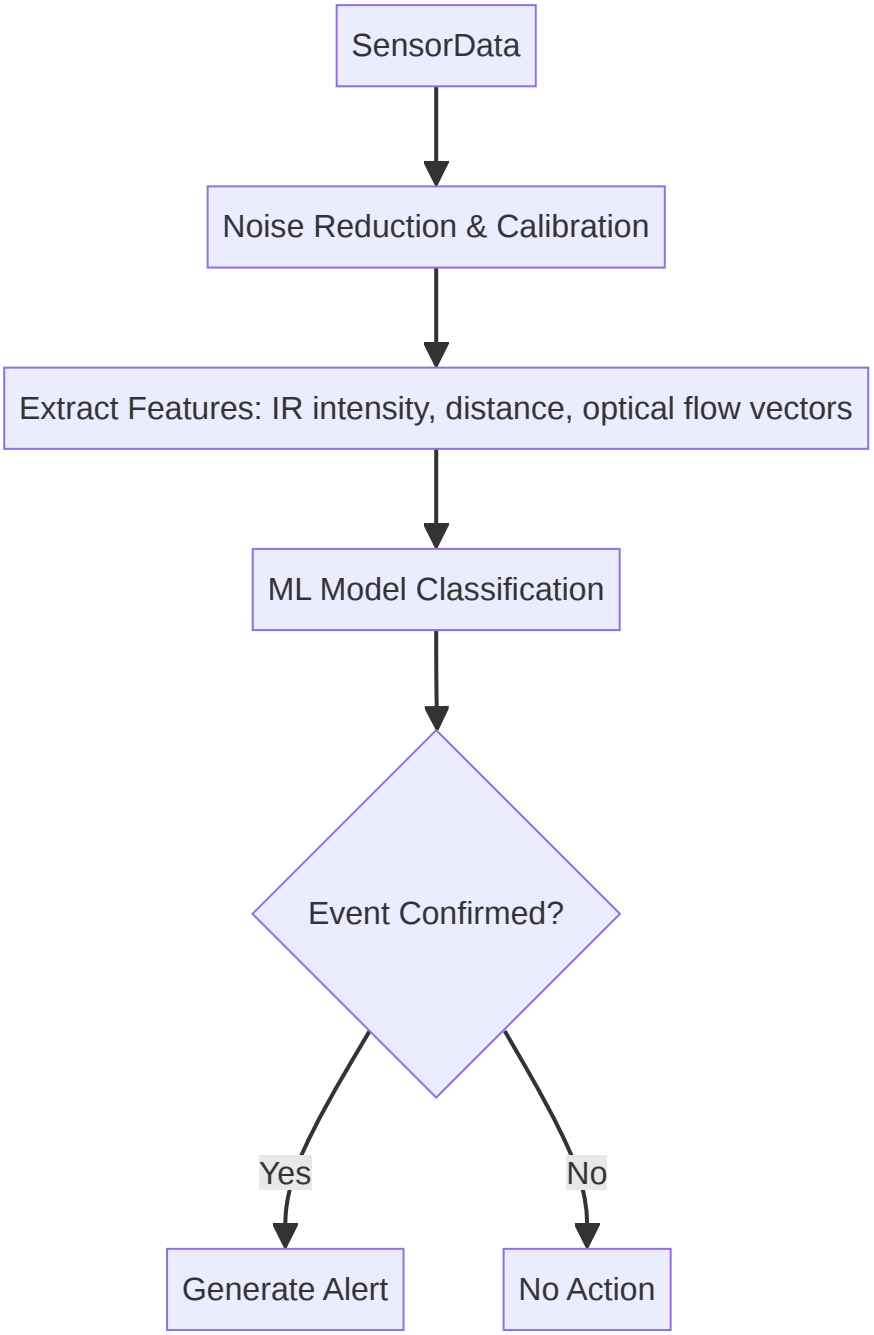
# 3. Technology Overview

## 3.1 Detection Algorithms

The core detection mechanism integrates multiple sensor inputs to achieve high confidence in identifying potential pool area intrusions while minimizing false positives from environmental noise (e.g., animals, weather). The system incorporates:

- **Sensor Fusion:** Combines infrared proximity data, ultrasonic distance measurements, and video-based optical flow analysis to enhance detection accuracy.
- **Signal Thresholding:** Adaptive thresholds are applied based on ambient conditions to differentiate between legitimate human presence and spurious signals.
- **Temporal Filtering:** Event persistence filters ensure transient noise does not trigger false alarms by requiring detection continuity over configurable time windows.
- **Machine Learning Models:** Supervised classification models (e.g., Random Forest, SVM) trained on labeled datasets distinguish between normal and anomalous pool area activities.

### Algorithmic Workflow:




## 3.2 Signal Processing Pipeline

1. **Raw Data Acquisition:** Sensor nodes sample data at 10–30 Hz frequency depending on sensor modality.
2. **Filtering:** Median and Kalman filters reduce random noise and smooth sensor outputs.

3. **Feature Extraction:** Key parameters such as proximity distance, motion vectors, and thermal gradients are computed.
4. **Normalization:** Feature vectors are normalized to a standard scale for consistent model input.
5. **Classification:** Processed features are input to the trained classifier to determine presence likelihood.
6. **Event Correlation:** Temporal and spatial correlation across multiple sensors reduces false alarms.

### 3.3 Sensor Hardware and Interfaces

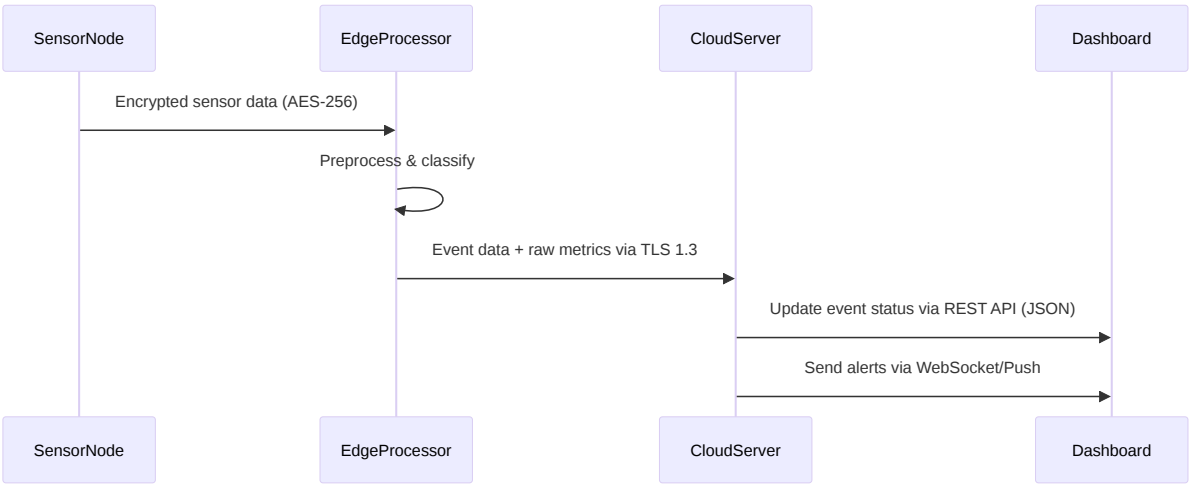
- **Infrared Sensors:** Passive infrared (PIR) sensors detect thermal emissions from human bodies within a 5-meter radius.
- **Ultrasonic Sensors:** Emit 40 kHz pulses and measure time-of-flight to detect object distance within a 10-meter range.
- **Optical Cameras:** Low-light cameras with embedded optical flow algorithms provide motion detection without recording personal visual data, thus preserving privacy.
- **Communication Interfaces:** Sensor nodes communicate via IEEE 802.11ax (WiFi 6) or low-power IEEE 802.15.4 (Zigbee) protocols depending on deployment scenario.

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## 4. Data Flow and Communication Protocols

Data is transmitted securely from sensor nodes to edge processors and cloud infrastructure using standardized protocols.

## Data Flow Diagram




## Communication Protocols

- **Link Layer:** WiFi 6 (IEEE 802.11ax) providing high throughput, low latency, and WPA3 security for sensor-edge communication.
- **Transport Layer:** TLS 1.3 for end-to-end encryption between edge processors and cloud servers.
- **Application Layer:** RESTful APIs over HTTPS for data queries and configuration; MQTT optional for low-latency event streaming.

## Data Formats

- Sensor data packets use a custom binary protocol encapsulated in JSON for cloud communication.
- Example JSON schema snippet for a detection event:

```
{
  "event_id": "123e4567-e89b-12d3-a456-426614174000",
  "timestamp": "2024-06-25T14:32:00Z",
  "sensor_node_id": "SN-01",
  "event_type": "intrusion_detected",
  "features": {
    "ir_intensity": 0.85,
    "distance_m": 2.3,
    "motion_vector": [0.12, -0.05]
  },
  "confidence_score": 0.97
}
```


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## 5. Software Components and Integration

The PAS software stack includes:

- **Embedded Firmware:** Real-time OS (FreeRTOS-based) managing sensor sampling, local processing, and wireless communication.
- **Edge Processing Software:** Linux-based application performing sensor fusion and initial classification using C++ and Python components.
- **Cloud Backend:** Microservices architecture deployed on AWS or Azure, running containerized services for data ingestion, model inference, and storage.
- **User Dashboard:** Web application built with React.js and secured with OAuth 2.0 authentication for role-based access control.

### Integration Methods

- **REST API:** Enables third-party systems to query sensor status, configure monitoring parameters, and retrieve historical data.
- **Webhooks:** Real-time notifications can be pushed to external systems upon event detection.
- **SDK:** Available for enterprise licensees to integrate PAS data streams into existing property management or retail analytics platforms.

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## 6. Security and Privacy Considerations

- **Data Encryption:** All data in transit and at rest is encrypted using AES-256 symmetric encryption and TLS 1.3 for communications.
- **Access Control:** Role-based access enforced on dashboards and APIs following Australian Privacy Principles to restrict data access.
- **Anonymization:** Optical sensors do not store or transmit personally identifiable video data; only motion vectors and metadata are processed.
- **Compliance:** System design adheres to the Privacy Act 1988 and relevant Australian consumer data protection laws.

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## 7. Technical Specifications

| Parameter | Specification | Notes |
|---|---|---|
| Sensor Sampling Rate | 10–30 Hz | Depending on sensor type |
| Detection Range | IR: up to 5 m | Ultrasonic: up to 10 m |
| Wireless Standards | IEEE 802.11ax (WiFi 6), IEEE 802.15.4 (Zigbee) | Configurable per deployment |
| Communication Security | TLS 1.3, WPA3, AES-256 | Industry-standard encryption |
| Edge Processor Hardware | ARM Cortex-A53 or better | 1.2 GHz minimum CPU frequency |
| Cloud Platform | AWS/Azure/GCP compatible | Containerized microservices architecture |
| Software Update Mechanism | OTA updates via secure channel | Signed firmware and software packages |
| Power Supply | 12 V DC (sensor nodes) | Optional solar/battery backup available |
| Operating Temperature | -10°C to 50°C | Suitable for outdoor installation |

# 8. Licensing Models and Terms

## Licensing Overview

Pool Area Shield is available for commercial licensing to qualified organisations within pool safety, property management, aged care, and retail analytics segments. Licensing ensures responsible use and protection of proprietary technology.

## Licensing Models

- **Per-Site License:** Annual fee per monitored pool location, includes software access, support, and updates.
- **Enterprise License:** Discounted volume pricing for organisations managing 10 or more sites.
- **OEM Integration:** White-label licensing allowing manufacturers to embed PAS technology into their hardware.
- **Reseller Program:** Authorised partnership agreements with industry stakeholders for distribution and support.

## Included in License

- Full access to PAS software suite and configuration tools.
- Detailed installation guides and training materials.
- Technical support during Australian business hours (AEST).
- Software updates and security patches for the license duration.
- Access to the monitoring dashboard and API endpoints.

## Exclusions

- WiFi or other network hardware (customers supply or purchase separately).
- On-site installation labor.
- Custom software integration or development beyond standard APIs.

- $^{24}/_{7}$ monitoring services (available as optional add-on).

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## 9. Compliance and Standards

Palatial Pool Services Pty Ltd ensures that Pool Area Shield technology complies with relevant Australian regulations and international standards, including:

- **Australian Standards AS 1926:** Specifies safety barriers and pool fencing requirements. PAS is a supplementary monitoring system and does not replace these requirements.
- **Privacy Act 1988 & Australian Privacy Principles (APPs):** Governs lawful collection, storage, and use of personal data. PAS anonymizes video data and restricts personal data usage strictly to safety monitoring.
- **IEEE 802.11ax (WiFi 6):** Wireless communication standard for sensor connectivity.
- **TLS 1.3 (RFC 8446):** Security protocol for encrypted communication.
- **AES-256 Encryption:** Industry standard for data encryption at rest and in transit.
- **ISO/IEC 27001 (Information Security Management):** PAS development processes align with best practices for secure software engineering and data management.

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## 10. Contact Information

For licensing enquiries, technical support, or further information regarding Pool Area Shield:

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